Integration of an Extended Kalman Filter into the SENTINEL-GNSS Project

Degradation-Aware GNSS / IMU Sensor Fusion for Urban Autonomous Driving

Team Pilot Project · Beihang University · June 2026

Why EKF for this Project?

- **Urban canyons, tunnels, multipath and NLOS break GNSS** — the exact conditions SENTINEL-GNSS targets.
- **Continuous navigation needs sensor fusion**: GNSS + IMU + odometry, not GNSS alone.
- **The motion model is nonlinear** — body-frame IMU accelerations rotate into the navigation frame by vehicle heading.
- **The EKF linearizes it with Jacobians**: accurate for our mild nonlinearity, fast enough for real time.
- **EKF over UKF for the pilot** — lower compute, simpler to validate; UKF kept as an advanced comparison.

Nonlinear state-space form

$$x_k = f(x_{k-1},\, u_k) + w_{k-1}$$
$$z_k = h(x_k) + v_k$$

The Project-Specific EKF Architecture

9-state vector

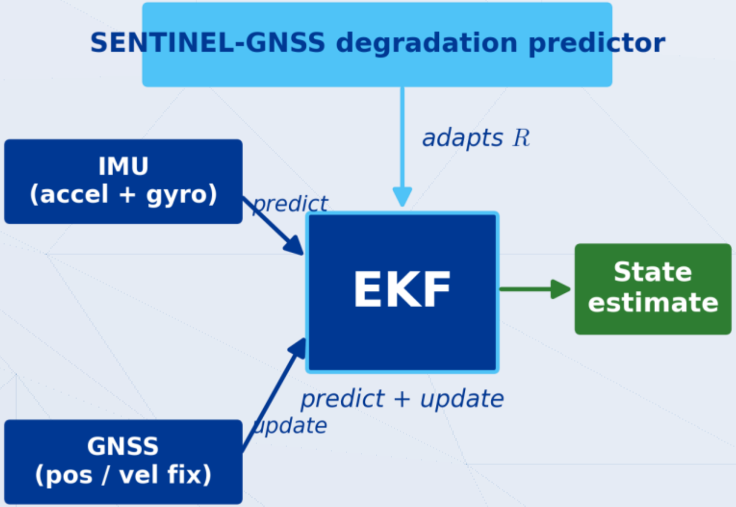
$$\mathbf{x} = [\, x,\ y,\ v_x,\ v_y,\ \psi,\ b,\ \dot{b},\ b_{a_x},\ b_{a_y}\,]^{\mathsf{T}}$$

- **Predict step** — driven by the IMU (accelerometer + yaw rate).
- **Update step** — GNSS position / velocity; linear measurement H.
- **Estimated online**: GNSS clock bias and drift, plus accelerometer biases.

Why "Extended"?

$$a^n = \mathbf{R}(\psi)\,(a^b - b_a)$$

body → nav rotation by heading makes f(·) nonlinear



Wiring the AI Prediction into the EKF

$$R_{\text{adaptive}} = R_{\text{base}}\,(1 + \alpha\, p_{\text{deg}})$$

The AI prediction enters the UPDATE step (via R) — not the predict step.

1. Take the calibrated P(DEGRADED) at the +5 s horizon from SENTINEL-GNSS.
2. Zero-order-hold the latest prediction to align predictor and filter rates.
3. Inflate the GNSS noise covariance R in proportion to it (hero formula).
4. Run the EKF update with the inflated R; tune scaling and threshold on validation.
5. If the prediction exceeds the threshold, coast on IMU (skip GNSS) for continuity.

EKF update with adaptive R

$$y_k = z_k - H\,\hat{x}_{k|k-1}$$

$$K_k = PH^\top\,(HPH^\top + R_{\text{adaptive}})^{-1}$$

Fallback — coasting

$$p_{\text{deg}} > p_{\text{thr}}$$

then coast on IMU, skip GNSS

Implementation Results & Expected Outcome

0.8206

+5 s Macro-F1

0.773

MCC (+5 s)

0.75 / 0.15

Tokyo DEGRADED F1
DL vs RF baseline

Expected fusion outcome

- Lower position RMSE during DEGRADED epochs vs a fixed-R baseline EKF.
- Maintained continuity — coasting holds the track through full GNSS loss.
- Degradation-aware trust: fewer multipath-driven position jumps.

Predictor metrics from Run 14 (current best). Fusion-stage RMSE pending — integration evaluation in progress.